

Autonomous Criminal Identification System

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Abstract—In an era of increasing concerns for public safety and surveillance, the ability to effectively detect and track human targets in public spaces is of paramount importance. This project focuses on the development of a robust system for human target detection, leveraging the state-of-the-art YOLOv8 Resnet 50 (You Only Look Once version 8) object detection model and DeepSORT (Deep Simple Online and Realtime Tracking) for tracking. This system is designed to operate in diverse public environments, ensuring the continuous monitoring of human activities. The YOLOv8 Resnet 50 model provides realtime object detection capabilities, allowing for precisely identifying human targets within public spaces. It is equipped to handle varying lighting conditions, making it suitable for both daytime and nighttime surveillance. DeepSORT complements the YOLOv8 Resnet 50 model by providing robust target tracking, ensuring that individuals can be monitored seamlessly as they move throughout the public area. The combination of YOLOv8 Resnet 50 and DeepSORT enables the system to send immediate alerts whenever anomalies or suspicious behavior are detected. This proactive approach can assist law enforcement agencies and security personnel in responding to potential threats quickly and efficiently. Additionally, the data generated by this system, including video footage and tracking information, can be stored for future reference and analysis, aiding in investigations and post-event reviews. The project's outcome presents a practical solution for enhancing public safety and security through advanced human target detection and tracking technology.

Index Terms— Human target detection, YOLOv8 Resnet 50, DeepSORT, public safety, surveillance, real-time tracking, object detection, anomaly detection, public places.

I. INTRODUCTION

In recent years, the need for effective surveillance and security measures in public spaces has become increasingly evident. The rising threats of criminal activities, acts of terrorism, and public safety concerns have underscored the importance of advanced surveillance systems. Traditionally, surveillance systems relied on human operators to monitor video feeds, which were often ineffective in crowded and dynamic public environments. We are motivated to prevent crimes like theft, vandalism, violence, and even acts of terrorism. We will use technology to watch what is happening in real-time and catch problems early. By doing this, we not only make public places safer but also discourage people from committing crimes, which makes everyone more secure. The primary objective of this project is to develop a resilient system that can effectively identify and monitor human subjects in various settings, functioning seamlessly during both day and night conditions. It is designed to recognize specific activities like running, and in the event of detecting any unusual behavior, the

system will promptly alert the relevant authority. This system operates by processing input from a camera and subjecting it to an anomaly detection algorithm. If the algorithm detects an anomaly, a visual alarm will be activated; otherwise, the frame will be ignored. Also, the ability to store and retrieve video footage and tracking data for future reference is valuable for post-incident investigations. This can aid in identifying suspects, understanding the sequence of events, and gathering evidence for legal purposes.

Section I: Introduction: Explains the problem and motivation for the project.

Section II: Existing system: Alike Systems which already exists and the drawbacks about it.

Section III: Implemented System: Explaining what is new in our project other than existing things, it includes the problem statement: Scope and hardware requirements.

Section IV: Conclusion and Future Scope.

A. Face Recognition for Criminal Identification: An Implementation of Principal Component Analysis for Face Recognition

The paper [1] discusses how CCTV footage can be used to identify criminals. The system consists of five stages: planning, requirement analysis, system design, implementation, and maintenance. The machine uses a database to display the person's personal information. They achieved 80 percent accuracy using the proposed model.

B. Criminal Identification System using Facial Recognition

The article [2] discusses the implementation of an automatic criminal identification system using face recognition technology. The system utilizes CCTV footage captured in public places to detect and recognize the faces of potential criminals. By comparing the captured images with a database of criminals if a match is found.

C. Airport Security using Face recognition

In this paper [3] Using the Passport database, they are identifying whether the traveler is an authorized passport holder or not.

D. Automatic Face Recognition Attendance System using Python and Opencv

The authors of this paper [4] are presenting an automatic face recognition system for attendance monitoring. They are capturing faces by using a camera and the captured image is compared with images that are already present in their database.

II. EXISTING SYSTEM

A. Manual Criminal Identification System

Manual identification systems rely on human effort and expertise to identify suspects and individuals of interest. Traditional methods include eyewitness accounts, composing sketches, and line-up identification. In manual systems, law enforcement officers or forensic experts play a significant role in the identification process. They may interview witnesses, create sketches, and assemble physical line-ups for eyewitness identification. Manual methods are often used when biometric data or automated systems are unavailable or inconclusive. Human Mistakes: Manual methods depend on what people see and remember, which can be influenced by their own opinions and biases. This can lead to mistakes. People can make errors when identifying suspects. Their memories can fade, they can be influenced by others, or they might have a harder time recognizing people from different racial backgrounds. Slower and Less Efficient: Manual methods take more time and effort. For example, organizing line-ups or sketching can be time-consuming. Expensive and Resource-Intensive: Manual methods require a lot of people, like police officers and forensic experts, which can be costly and not always available. Lack of Objective Data: Manual methods rely on descriptions and sketches, which can be less objective compared to using technology to analyze things like fingerprints or facial features. Ethical and Legal Issues: Manual methods can have problems with fairness and following the law, such as using suggestive methods or making mistakes. Inconsistent Results: Different people using manual methods might have different results, which can lead to problems with fairness and accuracy. Hard with Old Cases: For older cases with limited or outdated evidence, manual methods might not work well. Technology can sometimes help with these cases.

III. IMPLEMENTED SYSTEM

A. Problem Statement

The existing surveillance faces a range of challenges that badly affect public safety and security in crowded public places. Traditional surveillance methods, reliant on fixed cameras and manual monitoring, suffer from limited coverage, insufficient automation, and slow response to the detected event. These limitations not only compromise the security of individuals in these environments but also hinder the prevention of criminal activities, including theft, violence, and acts of terrorism.

B. Design of the System

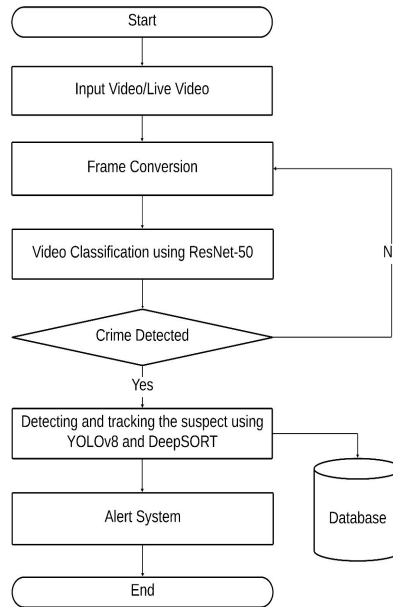


Fig. 1: Flow Chart

Description

1. The process starts with an Input Video/Live Video. 2. This video is then subjected to Frame Conversion. 3. The frames are processed by a Trained Model. 4. The model is responsible for Detecting Human presence in the frames. 5. If no human is detected, the process ends. 6. If a human is detected, the model proceeds to Crime Detection. 7. If a crime is detected, an Alert System is activated and the information is stored in a Database. 8. The process then ends. This flow chart describes the process of detecting crime in a video or live stream. It outlines how each frame of the video is analyzed by a trained model to detect human presence and potential criminal activity. If such activity is detected, an alert system is triggered and the information is stored for future reference. This could be a useful system for surveillance and security purposes.

C. Technology Stack

We have used YOLO v8 Resnet 50 as our object detection algorithm for the following reasons: 1. Improved Accuracy: YOLOv8 Resnet 50 has improved accuracy over its predecessors and other models, making it a reliable choice for object detection tasks. 2. Enhanced Speed: One of the most significant advantages of YOLOv8 Resnet 50 is its incredible speed. It can perform image segmentation at a rate of 81 frames per second, far exceeding the speed of other state-of-the-art models. 3. Multiple Backbones: This feature allows for flexibility and customization in model architecture, which can lead to better performance on specific tasks. 4. Adaptive Training: This feature allows the model to adjust its learning process based on the data it's trained on, potentially leading to better performance. 5. Advanced Data Augmentation: This can help improve the model's performance by providing more varied training data. 6. Customizable Architecture: This allows you to tailor the model to your specific needs, potentially improving performance on your specific task. 7. Pre-Trained Models: These can save you time and computational resources as you don't need to train the model from scratch.

System Architecture

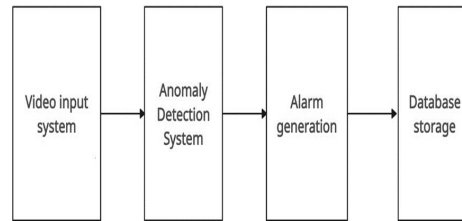


Fig. 2: Block Diagram

Description

1. Video Input System: The system starts with one or more cameras that provide live video feeds as input. These cameras capture the surrounding environment and send the video stream for further processing. 2. Video Feed Preprocessing: The video feed preprocessing stage is essential for improving the quality and preparing the video for subsequent analysis. It encompasses three main components. First, "Frame Capture" captures individual frames from the video feed. Next, "Image Enhancement" applies various filters and adjustments to enhance the image quality. Finally, "Frame Differencing" compares consecutive frames to detect changes, particularly focusing on motion detection. 3. Anomaly Detection System: The "Anomaly Detection" system plays a crucial role in analyzing the pre-processed video frames to identify any anomalous activities. It involves three main components. "Object Detection" identifies objects or people within the video frames. "Behaviour Analysis" then examines these detected objects' behavior and movement patterns. The "Anomaly Classifier" determines if the observed activities deviate from expected norms, based on predefined criteria. 4. Alarm Generation: When the Anomaly Detection system detects an anomaly, the "Alarm Triggering" stage comes into play. Its function is to generate an alarm when unusual activities are identified. The key components include "Alarm Generation," which generates an alarm signal or message, and "Alert Notification," responsible for notifying the relevant parties or systems about the detected anomaly. 5. Database Storage: The system may incorporate a "Database" component, which serves multiple functions. It stores system configurations and user profiles in the "Configuration Database." Additionally, the "Historical Data" repository is used to store historical data for subsequent analysis and reporting. This database is optional and may be implemented to maintain records of system-related information.

D. Step-By-Step Working

```
Microsoft Windows [Version 10.0.22631.3296]
(c) Microsoft Corporation. All rights reserved.

C:\Users\chand\Desktop\rajpofinal_single>python manage.py runserver
```

Fig. 3: Move to the correct directory where the files are present and start Django server

```
C:\Users\chand\Desktop\rajpofinal_single>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

Neither CUDA nor MPS are available - defaulting to CPU. Note: This module is much faster with a GPU.
C:\Python311\Lib\site-packages\numpy\core\numeric.py:2463: FutureWarning:
elementwise comparison failed; returning scalar instead, but in the future
will perform elementwise comparison
  return bool(asarray(a1 == a2).all())
System check identified no issues (0 silenced).
April 04, 2024 - 11:24:24
Django version 4.2, using settings 'rajpofinal.settings'
Starting development server at http://127.0.0.1:8000/
```

Fig. 4: Click on the link generated

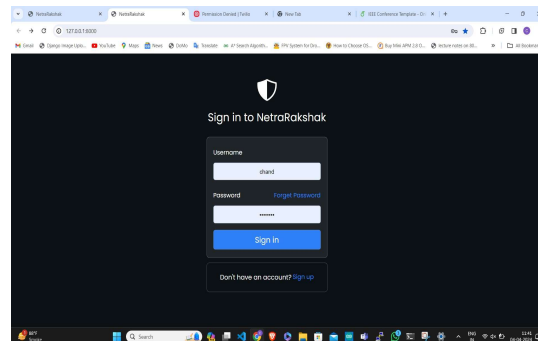


Fig 5. Login page of website multaneously

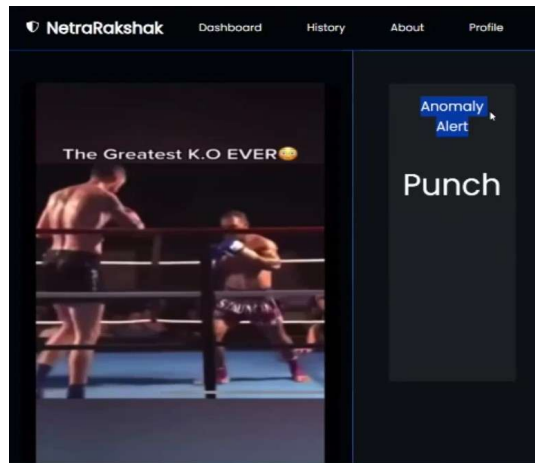


Fig. 6: RESNET50 model detected action as punch with accuracy greater than 95%

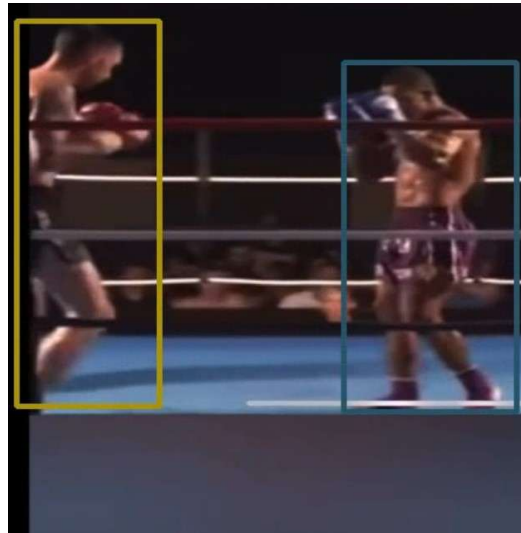


Fig. 7: YOLOv8+deepsort+easyocr all working siFig

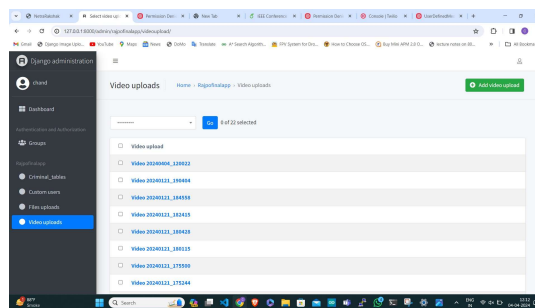


Fig. 8: Result stored in database as per date and time

IV. CONCLUSION

In conclusion, this project addresses a pressing need for enhanced surveillance and security measures in today's public spaces. The increasing concerns related to criminal activities, acts of terrorism, and overall public safety undermine the urgency of developing advanced surveillance systems. Traditionally, relying solely on human

operators for video monitoring has proven inadequate, particularly in crowded environments. By using technology, we aim to not only respond effectively to security concerns but also to deter potential criminal activities, ultimately making public spaces safer and more secure for everyone.

FUTURE SCOPE

The system's criminal activity detection is hindered by unclear CCTV video during adverse weather conditions. Environmental challenges may limit the accurate identification and monitoring of criminal activities, affecting system performance in enhancing public safety. The system may lose track of a target if a sizable obstruction, such as a large object, interrupts the tracking path. Instances of losing the target due to obstructions can occur, potentially affecting continuous and seamless tracking within public spaces.

Our future goal is to connect our software directly to live CCTV feeds, allowing real-time human target detection and tracking for enhanced public safety measures. Developing a website that allows authorized personnel to access the surveillance system remotely can significantly enhance its usability. This can enable law enforcement and security personnel to respond quickly to alerts and monitor situations from anywhere.

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